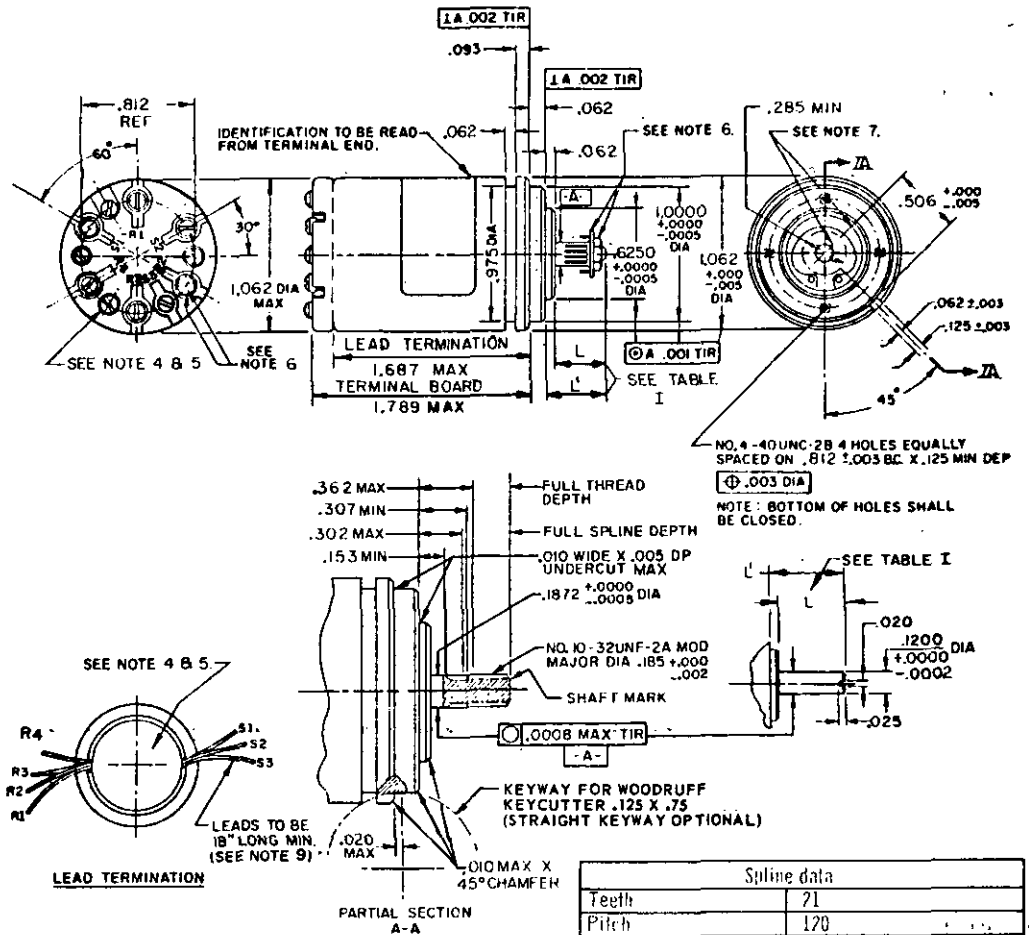
MILITARY SPECIFICATION SHEET

TRANSOLVER
TYPE 90 11S210Y440A

The complete requirements for procuring the transolver described herein shall consist of this document and the latest issue of MIL-T-83727.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Spline data	
Teeth	21
Pitch	120
Pitch diameter	.175 +.000, -.002
Outside diameter	.1872 +.0000; -.0005
Pressure angle	20°
Root diameter maximum	.155
Tooth form	Full depth involute
Note: The end function of this spline is for positive rotational fastenings. It is not to be used as a gear. This spline shall be inspected by (spline) go composite ring gage MS 90406.	

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Recess shall allow the terminal lugs to be installed with the barrels inboard or outboard, and prevent the lugs from turning. Terminal block designed for MS17182 terminal lug. Terminal lead wires shall depart through the back surface of the terminal end of the transolver in any acceptable manner.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. The following list of hardware shall be packaged in a ventilated envelope made of material having a pH value of 6-8 and compatible with the weight of the contents. The packaged hardware shall be placed with the unit transolver as follows:

Machine screw	(MS 35276-203)	6 Required (*) (**)
Lockwasher	(MS 35338-134)	6 Required (*) (**)
Drive washer	(MS 17186-6)	1 Required (*)
Nut	(MS 17187-2)	1 Required (*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.

7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. transolver zero, and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the transolvers to be furnished under this specification shall be subjected to the first article tests as specified in MIL-T-83727.
9. Leads shall be type E-28.

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REQUIREMENTS:	VALUE	UNIT	TOLERANCE
Primary element	Stator	-	-
Excitation frequency	400	Hz	±1%
Primary voltage	90	Volts	P_{I} ±1%
Primary current	42	Milliamperes	Max.
Primary power	.43	Watts	Nominal
D.C. resistance:			
Stator	315	Ohms	±15%
Rotor	80	Ohms	±15%
Impedance:			
Z_{so}	333+j2033	Ohms	R±15%+jx±10%
Z_{ro}	95+j344	Ohms	R±15%+jx±10%
Z_{rs}	113+j33	Ohms	Referenced
Transformation ratio			
Output/input	.335	-	±2%
Phase shift	6.3	Degrees	±1.5°
Electrical error	6	Minutes	Max.
Interaxis error	4	Minutes	Max.
Null voltage:			
Fundamental	40	MV rms	Max.
Total	60	MV rms	Max.
Friction torque:			
At 23° C	.07 (5)	Oz-in. (Gm-Cm)	Max.
At -55° C and +125° C	.15 (11)	Oz-in. (Gm-Cm)	Max.
Radial play	.0006 (Using 4 oz. load)	Inches	Max.
End play	.0001-.0015 (Using 8 oz. load)	Inches	Min.-Max.
Temperature rise	5	°C	Max.

Part number: M83727/16 and a dash number from table I.

TABLE I. Dash number and characteristics.

Dash no.	Shaft diameter	Shaft style	Termination	Shaft length	
				L ref	L' dim to first mounting surface (±.010)
-001	.1872	Splined	Terminal board	.500	.562
-002	.1200	Plain	Wire leads	.500	.562
-003	.1200	Plain	Terminal board	.500	.562

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Custodians:
Navy - AS
Air Force - 80

Review activities:
Navy - SH, EC
Air Force - 11, 17
DSA - ES

User activities:
Navy - CG, MC
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0272)

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